

TL1 – Class Diagram & Questions

1. What design pattern is being used?

The State design pattern is used to manage the different states of the game. GameState defines the common behavior for all states, while MenuState, PlayingState, PausedState, and GameStateManager maintains the current state and controls transitions between them.

2. How is dynamic binding used?

Dynamic binding occurs when GameStateManager calls methods such as Enter() and Update() through a GameState reference. At runtime, the overridden method belonging to the current state is executed. For example, if the current state is PausedState, calling Enter() executes the PausedState implementation rather than the base GameState implementation.

Class Diagram: UI/HUD, Game State & Menu System (TL1)

